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Adaptive scheduling strategy of fog computing tasks with different priority for intelligent production lines

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Abstract

Fog computing has been widely deployed in intelligent production lines to provide real-time computing services for terminal devices to alleviate the transmission problem between cloud and terminal. Nevertheless, when there are many services provided in a single fog node, how to choose an optimal data processing path is a challenge to minimize optimization latency and power consumption while ensuring the reliable transmission of data with different priorities. To address the problems as mentioned above in this paper, we first propose a joint evaluation model of time latency and power consumption based on task priority. Then, we formulate a fog computing adaptive scheduling algorithm (FCAS) based on dynamic programming to obtain the optimal path for processing data with different priorities at fog nodes. Experimental results illustrate that our proposed evaluation model and algorithm have lower power consumption, high efficiency, and higher reliability.

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Keywords: Intelligent production line; Fog node; Adaptive scheduling; Multi-objective optimization; Dynamic programming algorithm;

1. Introduction

Manufacturing is a momentous economic pillar industry of a country, and the advanced degree of manufacturing technology represents the level of productivity of a country. Throughout the world, countries have made every endeavor to invest in the field of manufacturing technology. For example, Germany proposed the "Industry 4.0" strategy, the United States put forward the "Industrial Internet", and China formulated the "Made in China 2025" strategy [1]. In recent years, the development of emerging information technologies such as the Internet of Things,

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big data, and cloud computing has promoted the transformation of manufacturing technology, leading the transformation and upgrading of traditional manufacturing to intelligent manufacturing [2].

With the application of the Internet of Things technology in intelligent factories, various terminal devices such as production equipment, transmission equipment, and sensing equipment have realized the connection based on a variety of heterogeneous communication devices. The traditional information isolated island is evolving towards the mode of interconnection, so the informatization level of manufacturing systems is gradually improving. Compared with the single production of traditional manufacturing, intelligent manufacturing needs to meet the market demand for personalized production, which promotes the continuous reorganization of production lines and the continuous upgrading of processes. At the same time, it puts forward higher requirements on the real-time, energy efficiency, and reliability of the computing system on which it depends.

We first propose a joint evaluation model of time delay and power consumption based on task priority and then propose a self-assessment model based on fog computing, which solves the problems of weak computing power, high latency and high-power consumption of fog nodes in smart production lines. The adaptive scheduling algorithm planned to obtain the best path to process data with different priorities at the fog node.

2. Related Work

Cloud computing [3] provides rich computing, storage and other application services for intelligent manufacturing. However, the intelligent production line system based on cloud computing still faces many unsolved problems. Suppose cloud computing continues to be used to process massive manufacturing data. In that case, the long-distance transmission of massive manufacturing data will not only lead to high latency and network congestion [4] but also significantly reduce the security of manufacturing data in the long-distance transmission process.

Fog computing [5] is an emerging technology proposed to make up for the deficiencies of cloud computing. As an expansion and promotion of cloud computing, fog computing migrates cloud computing services to the edge of the network and builds a fog computing platform by deploying a certain number of fog nodes near terminal devices or data sources. Therefore, in order to meet the high-performance requirements of the task, the intelligent terminal equipment can select fog nodes to complete the task processing. In recent years, fog computing applied in many industrial scenarios, such as intelligent factories, smart grids and internet of vehicles.

Fog computing task scheduling has been a prevalent open problem in fog computing that captures the interests of a significant number of scholars. To illustrate, Zhang et al. [6] proposed a task fair offload analysis framework, which first formulated the latency and energy consumption, and then established a fair scheduling mechanism for each fog node. Liu et al. [7] studied the task scheduling in the fog computing micro datacenter networks, designed the mathematical model of the completion time of a single job, and reduced the completion time of a single job through heuristic algorithm. Yang et al. [8] studied the balance between latency and energy consumption in task scheduling in heterogeneous fog networks. Pham et al. [9] studied the task scheduling problem in cloud-fog computing, established a mathematical model of task scheduling, and realized the balance of completion time and cost by introducing heuristic algorithm. However, the algorithms, as mentioned above, seem hard to be applied in our problem due to the complexity of constraints. We should propose a more effective solution formation strategy to ensure feasibility.

3. Fog computing task scheduling

3.1. Task scheduling description

In the intelligent production line, a series of sensors at the terminal upload the data set $\Gamma = \{1, 2, 3, \dots, N\}$ that needs to be processed. Each data processing needs to involve the function set $\Omega_\gamma (\gamma \in \Gamma, \Omega_\gamma \subseteq S)$ in the fog node, and $S = \{1, 2, \dots, M\}$ denotes the set of all functions contained in the fog node. When the data flow enters the fog nodes, different data flows will bring different impacts. At the same time, in order to ensure the timely processing of data with high priority and reduce the power consumption of fog nodes as much as possible. Therefore, it is necessary to design a more efficient scheduling algorithm to realize joint scheduling to maximize performance indicators. For

example, in Fig. 1, the same service exists in the fog node, and data processing has multiple paths. Therefore, it is necessary to find a path that can maximize performance in fog nodes through a joint scheduling algorithm.

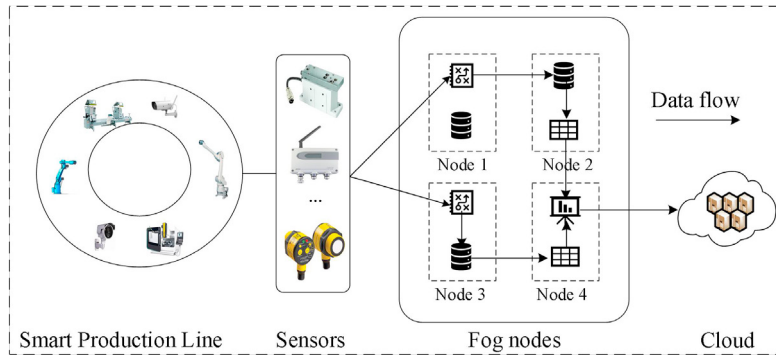


Fig. 1. Task scheduling problem description of intelligent production line

3.2. Objective optimization problem description

The data processing flow in the fog node shown in Fig. 2. We assume that when processing data in the fog node, there is a chain topology between various service functions. The data processing process is a linear calculation process similar to pipeline mode. In this process, the current service starts to perform calculation after receiving data from other preamble services.

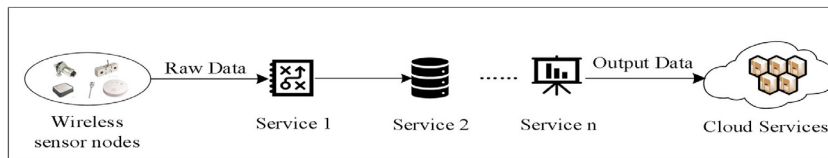


Fig. 2. Data processing flow in the fog nodes

4. Task scheduling optimization problem

4.1. Objective optimization problem description

Aiming to ensure that the path chosen by the data flow in the fog node is the optimal path, this paper takes the latency and power consumption as the targets to solve the problem and find a service path that can satisfy the performance maximization of the fog node. The objective optimization problem will detail in the next.

We let $\Gamma = \{1, 2, 3, \dots, N\}$ to denotes the data set to be processed uploaded by a series of sensors, where the processing of each data needs to involve the service set in the fog node as $\Omega_\gamma (\gamma \in \Gamma, \Omega_\gamma \subseteq T)$, and $S = \{1, 2, \dots, M\}$ denotes the service set contained in the fog node. When data enters the fog meter node network, different data flows will have different impacts. Therefore, it is necessary to design a more efficient scheduling algorithm to realize joint scheduling in order to obtain the service path with optimal latency and power consumption in fog nodes.

4.2. Formulation of objective optimization problem

(1) Node task description

Fog node set for $I = \{1, 2, \dots, P\}$, $X_{i,j}$ is a binary variable, indicating whether service j is provided in fog node i , so:

$$X_{i,j} = \begin{cases} 1, & \text{if server } j \text{ exists on node } j \\ 0, & \text{others} \end{cases} \quad (1)$$

With the purpose of each fog node can fully play its role, each node should have at least one service, so we have:

$$\sum_{j \in T} X_{i,j} \geq 1 \quad \forall i \in I \quad (2)$$

(2) Data flow on chain service model

In the chained service model, to represent the direction of data flow, the binary variable $\Phi_{\gamma,i,j}^j$ ($\gamma \in \Gamma, i \in \Omega_\gamma, j \in I$) is introduced, and $\Phi_{\gamma,i,j}^j = 1$ means that when data s is processed by the i service in fog node j , the flow direction of data flows from j to j' . Since data can only flow to one node per hop, there are the following constraints:

$$\sum_{j' \in I} \Phi_{\gamma,i,j}^j = 1 \quad (3)$$

We assume that any fog node can process the data collected by the sensor node, so for all $\gamma \in \Gamma, j \in I$, we should have:

$$\Phi_{\gamma,1,0}^j = 1 \quad (4)$$

(3) Service latency description

After introducing several constraints, we are now ready to introduce the objective function formally. The metrics we are concerned about are the transmission time, waiting time and data processing time. Formally, we let $D_{j,j'}$ ($j \in I$) denotes the distance between fog nodes, $Tr_{j,j'}$ ($j \in I$) denotes the latency rate of transmission between fog nodes, and $W_{\gamma,i,j}$ ($\gamma \in \Gamma, i \in \Omega_\gamma, j \in I$) denotes the wait time of fog node j providing the service i . $Del_{\gamma,i,j}$ ($\gamma \in \Gamma, i \in \Omega_\gamma, j \in I$) denotes the time when the service i in fog node j processes, namely, we have the following formulation:

$$\tau_{\gamma,k} = \sum_{i \in \Omega_\gamma} \sum_{j \in I} \sum_{j' \in I} \Phi_{\gamma,i,j}^j \times (D_{j,j'} \times Tr_{j,j'} + Del_{\gamma,i,j} + W_{\gamma,i,j}) \quad (5)$$

Where, $\gamma \in \Gamma, k \in P, P = \{1, 2, \dots, 0\}$ denote the set of candidate paths.

(4) Service power description

The metrics of service power we are concerned about are the transmission power and data processing power. Formally, we let $D_{j,j'}$ ($j \in I$) denotes the distance between fog nodes, $P_{j,j'}$ ($j \in I$) denotes the latency rate of transmission between fog nodes and $DelPower_{\gamma,i,j}$ ($\gamma \in \Gamma, i \in \Omega_\gamma, j \in I$) denotes the power consumption of data processing by the service i in fog node j . Then, we have the following formulation:

$$\pi_{\gamma,k} = \sum_{i \in \Omega_\gamma} \sum_{j \in I} \sum_{j' \in I} \Phi_{\gamma,i,j}^j \times (Dis_{j,j'} \times P_{j,j'} + DelPower_{\gamma,i,j}) \quad (6)$$

$$\gamma \in \Gamma, k \in P \quad (7)$$

(5) Latency and power evaluation model based on task priority

In the intelligent production line, because different fog nodes provide the same service, there are many choices for data flow between fog nodes, thus bringing different system overhead. At the same time, tasks have different priorities. If only the latency or power consumption of the system considered, the requirements of low latency for high-priority events and low power consumption for low-priority events cannot meet.

In consideration of the above problems, we propose a task priority-based latency and power evaluation model, which is an adaptive adjustment strategy of latency and power realized according to different data priorities. While

ensuring high priority data processing with low latency, the overall power consumption of the system reduced. The evaluation model is as follows:

$$\eta_\gamma = \text{Max}_{i \in P} \left(\frac{1}{(1+\tau_{\gamma,i})^\lambda + (1+\pi_{\gamma,i})^{3-\lambda}} \right) \quad (8)$$

$$\lambda = \begin{cases} 1, & \text{high-priority tasks} \\ 2, & \text{lower-priority tasks} \end{cases} \quad (9)$$

$$\tau_i \leq \text{Max}_{\text{Latency}}, \pi_i \leq \text{Max}_{\text{Power}}, \forall i \in P \quad (10)$$

In the formula (8), $\eta_\gamma (\gamma \in \Gamma)$ denotes the score of the path selected during processing of the γ th data, λ denotes the priority of each data, τ_i denotes the time spent in processing the data in the i path, π_i denotes the power consumed in processing the data in the i path, $\text{Max}_{\text{Latency}}$ denotes the maximum tolerated latency of the system, and $\text{Max}_{\text{Power}}$ denotes the maximum tolerated power of the system.

5. Fog computation adaptive scheduling algorithm

Dynamic programming algorithm is best known for solving problems with optimal properties [10]. By decomposing the problem to be solved into several sub-problems, the solution of the original problem obtained by solving the solution of the sub-problems. The decision-making process of the dynamic programming algorithm shown in Fig. 3. In this paper, the latency and power consumption of data under different priorities are evaluated through formula (8) to solve the problem of optimal path selection in fog nodes.

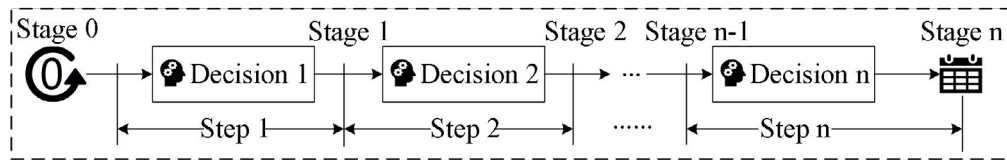


Fig. 3. Decision-making process of dynamic programming algorithm

(1) Algorithm description

We select the sensor node as the starting point and use the evaluation model (8) based on the priority of the task to initialize the resource consumption of all fog nodes connected to the sensor node when processing tasks. Moreover, taking the minimum resource consumption of each fog node processing task as the decision condition, calculate the corresponding minimum resource consumption of all nodes when completing different tasks in turn, and finally obtain the optimal path with the least resource consumption. Finally, the optimal path with the least resource consumption obtained. Then, we have the following state transition equation:

$$f_{i,j+1} = \arg \min \{f_{k,j} + \text{transfer}_{k,i} + \text{finish}_{i,j+1}\} \quad (11)$$

$$i, k \in I, \quad j \in S \quad (12)$$

Where, $f_{i,j}$ denotes the corresponding resource consumption when the service j is completed in the fog node i . $\text{transfer}_{k,i}$ denotes the resource consumption caused by $k \rightarrow i$ transmission of fog node. $\text{finish}_{i,j+1}$ denotes the resource consumption caused when the service $j+1$ is completed in the node i .

(2) Algorithm implementation

The pseudo-code of our proposed FCAS algorithm shown in Algorithm 1. Firstly, the parameters in the algorithm are input and initialized. The primary process of the algorithm is in lines 3-8. By enumerating nodes and services

(lines 1-2), it judged whether service j exists in i (line 3), and then enumerating node k (line 4) of task $j-1$ before fog node i process task j . Lines 5-8 update formula (11) by using the data calculated by the formula (8) and the optimal solution has obtained.

Algorithm 1

Initialize: $\lambda, f_{0,j}, \Gamma, \Omega_{\gamma}, S, I, D, Tr, Del, P, W, DealPower$

```

01 for  $i = 1; i \leq N; i++$  do
02   for  $j = 2; j \leq S$  do
03     Select a next server  $j$  according to  $X_{i,j}$ 
04     for  $k = 1; k \leq N; k++$  do
05       Select a next node  $k$  according to  $\Phi_{\gamma,i,j}^k$ 
06       get front node  $k$ 
07       Calculate the value of the next node  $k$  according to (8)
08       Select the minimum of  $f_{i,j+1}$  according to (11)
09     end
10   end
11 end

```

6. Simulation and results analysis

In this section, we use the iCandyBox intelligent production line as a model to simulate the task distribution in the fog node when the whole production line is running by controlling the k parameter. We randomly generate delay parameters between [1.0ms, 9.9ms] and randomly generate power consumption parameters between [1mW, 9.9mW]. According to the above method, 40 sets of test data selected for experimental verification, each set of data run five times, and the average of the obtained results used as the comparison data. We compare FCAS with MRPACO [14], and the comparison results will introduce next.

The comparison between Figs. 4 and 5 show that FCAS has a small improvement in latency, which can reduce the latency by 2.7% ~ 27.9%, but save 22.2% ~ 67.2% in power consumption. It is because MRPACO algorithm takes time as the optimization parameter. For high-priority data, the results obtained by the two algorithms are similar. However, for low-priority data, FCAS can better converge in the direction of lower power consumption. Therefore, compared with MRPACO algorithm, FCAS saves processing time and dramatically reduces the power consumption of fog nodes in processing data. For some data, the latency of FCAS is higher than that of MRPACO algorithm, which is also because FCAS considers the power consumption of low-priority data, comprehensively evaluates the latency and power consumption, and finally sacrifices part of the latency to obtain lower power.

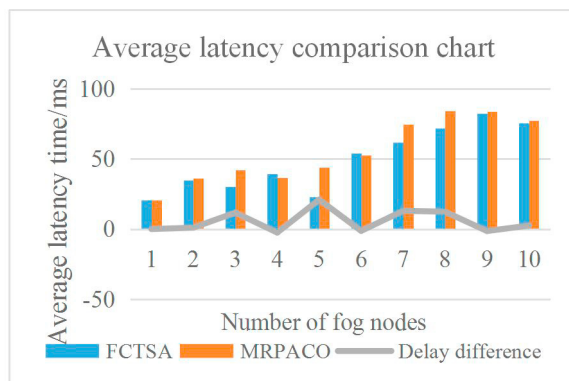


Fig. 4. Comparison of Average Latency

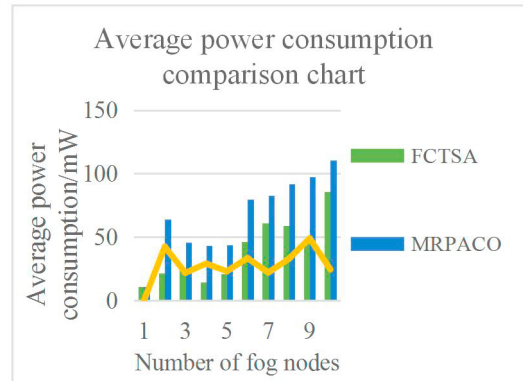


Fig. 5. Comparison of Average Power Consumption

7. Conclusion

In this paper, we aim to solve the problems of high latency and large power consumption in iCandyBox intelligent production line. We propose a joint evaluation model of latency and power consumption based on task priority and formulate an adaptive scheduling algorithm for fog computing based on dynamic programming. It solves the problem of adaptive path selection for data streams with different priorities in fog nodes and realizes low latency transmission of high priority data and low power transmission of low priority data. The experimental results illustrate that the time latency not only reduced by 2.7% ~ 27.9% but also the power consumption is saved by 22.2% ~ 67.2% through the fog node task adaptive scheduling optimization algorithm. The evaluation model and algorithm proposed in this paper not merely ensure the stability of data transmission, but also have an excellent energy-saving effect.

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